

D. Shocking Arrangement

time limit per test: 2 seconds
memory limit per test: 256 megabytes

You are given an array $a_1, a_2, \dots, a_n$ consisting of integers such that $a_1 + a_2 + \dots + a_n = 0$.

You have to rearrange the elements of the array a so that the following condition is satisfied:

$$\max_{1 \leq l \leq r \leq n} |a_l + a_{l+1} + \dots + a_r| < \max(a_1, a_2, \dots, a_n) - \min(a_1, a_2, \dots, a_n),$$

where $|x|$ denotes the absolute value of x .

More formally, determine if there exists a permutation $p_1, p_2, \dots, p_n$ that for the array $a_{p_1}, a_{p_2}, \dots, a_{p_n}$, the condition above is satisfied, and find the corresponding array.

Recall that the array $p_1, p_2, \dots, p_n$ is called a permutation if for each integer x from 1 to n there is exactly one i from 1 to n such that $p_i = x$.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 50\,000$). The description of the test cases follows.

The first line of each test case contains a single integer n ($1 \leq n \leq 300\,000$) — the length of the array a .

The second line of each test case contains n integers $a_1, a_2, \dots, a_n$ ($-10^9 \leq a_i \leq 10^9$) — elements of the array a . It is guaranteed that the sum of the array a is zero, in other words:
 $a_1 + a_2 + \dots + a_n = 0$.

It is guaranteed that the sum of n over all test cases does not exceed 300 000.

Output

For each test case, if it is impossible to rearrange the elements of the array a in the required way, print "No" in a single line.

If possible, print "Yes" in the first line, and then in a separate line n numbers — elements $a_1, a_2, \dots, a_n$ rearranged in a valid order ($a_{p_1}, a_{p_2}, \dots, a_{p_n}$).

If there are several possible answers, you can output any of them.

Example

input	output
7	Yes
4	-5 -2 3 4
3 4 -2 -5	Yes
5	-3 2 -3 2 2
2 2 2 -3 -3	Yes
8	1 1 1 -3 1 1 1 -3
-3 -3 1 1 1 1 1 1	Yes
3	-1 0 1
0 1 -1	Yes
7	4 -4 4 -4 0 3 -3
-3 4 3 4 -4 -4 0	No
1	Yes
0	13 12 -18 15 -18 13 -17
7	
-18 13 -18 -17 12 15 13	

Note

In the first test case $\max(a_1, \dots, a_n) - \min(a_1, \dots, a_n) = 9$. Therefore, the elements can be rearranged as $[-5, -2, 3, 4]$. It is easy to see that for such an arrangement $|a_l + \dots + a_r|$ is always not greater than 7, and therefore less than 9.

In the second test case you can rearrange the elements of the array as $[-3, 2, -3, 2, 2]$. Then the maximum modulus of the sum will be reached on the subarray $[-3, 2, -3]$, and will be equal to $|-3 + 2 + -3| = |-4| = 4$, which is less than 5.

In the fourth test example, any rearrangement of the array a will be suitable as an answer, including $[-1, 0, 1]$.